

Kazi Md. Tawsif Rahman

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Research Summary

Computer Science graduate and backend systems engineer investigating privacy risk in machine-learning-based network intrusion detection. My current independent study uses a locked IDS evaluation protocol and shadow-model membership-inference auditing; one formally accounted DP-SGD feasibility run is complete, and the full privacy-budget comparison is in progress.

Research Interests

Network security; privacy-preserving machine learning; membership-inference auditing; trustworthy evaluation of security-critical ML; reproducible experimental systems.

Education

Chittagong University of Engineering and Technology (CUET), Bangladesh Sep 2024
B.Sc. in Computer Science and Engineering CGPA: 3.59 / 4.00

Research Experience

A Reproducible Privacy–Utility Audit for ML-Based Network Intrusion Detection

Independent Research Project 2026 – Present

- Designed a locked NSL-KDD binary-classification protocol with a 70/10/20 target-train/validation/shadow split, validation-only threshold selection, and untouched KDDTest+ utility evaluation.
- Established archived Random Forest, XGBoost, and MLP IDS baselines. In the pre-MIA comparison, the validation-selected MLP threshold of 0.28 produced 76.91% Recall, 23.09% FNR, and 84.08% F1 on KDDTest+.
- Built five shadow MLPs and evaluated score-only black-box and label-aware audit threat models. The strongest overall held-out attack remained near chance ($\text{ROC-AUC} \approx 0.503$), so I treat the floor effect as a limitation, not as evidence that the model is private.
- Rebuilt the target MLP in PyTorch and completed one Opacus DP-SGD feasibility run with explicit accounting ($\epsilon \approx 8.0$, $\delta = 1.13 \times 10^{-5}$). It achieved 66.20% Recall, 33.80% FNR, 76.80% F1, and 91.93% PR-AUC; this run validates the implementation, not reduced membership leakage.
- In progress: privacy-budget sweep, the same fixed membership audit for each DP condition, bootstrap uncertainty, and repeated runs for selected settings.

Code and artifacts: github.com/tawsif113/privacy-utility-dp-ids

Publications

- Kazi Md. Tawsif Rahman** and Chowdhury Mahfuzul Hoq. “An Automated System for Detecting Property Insurance Fraud Using Machine Learning.” *International Journal of Mathematical Sciences and Computing*, 2024. doi:10.5815/ijmsc.2024.03.02.
- Mahfuzulhoq Chowdhury, **Kazi Md. Tawsif Rahman**, and Hossain Ahmad Maruf. “Whistle Blower: An Insurance Awareness Mobile Application with Insurance Policy Selection, Fraud Detection, Critical Help, Complaint Features.” *2024 8th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)*, pp. 1–6, 2024. doi:10.1109/CSITSS64042.2024.10817002.

Selected Professional Experience

Software Engineer, BRAC IT Services Ltd., Dhaka, Bangladesh

Dec 2024 – Present

CRM Platform

- Built configurable rule-based routing with priorities, schedules, conditional matching, rotation rules, duplicate checks, and validated field/operator/value conditions.
- Replaced full-column Kanban reordering with fixed-length order keys and reduced each movement to a single-rank assignment.
- Added access controls, idempotent moves, transition validation, audit logs, and timeline entries to make state changes traceable.

Ticketing System

- Built services for ticket creation, search, comments, forwarding, assignment rules, teams, SLA/KPI tracking, feedback, notifications, and metadata.
- Implemented least-loaded team routing with Redis-backed workload counters and maintained PostgreSQL–Redis workload consistency after ticket creation, reassignment, and resolution.

Technical and Research Skills

Research methods Locked-split experimental design, validation-only threshold selection, shadow-model MIA, privacy accounting, bootstrap uncertainty, reproducibility manifests, threat-model specification

ML / Privacy Python, scikit-learn, XGBoost, PyTorch, Opacus, DP-SGD, model evaluation

Backend / Data Java, Spring Boot, Spring Security, JPA/Hibernate, REST APIs, PostgreSQL, MongoDB, Redis, RabbitMQ, Flyway

Engineering SQL, C, C++, Docker, Linux, Git, JUnit 5, Mockito, API testing

Awards and Achievements

- Codeforces Specialist; rating above 1500.
- Finalist, CUET IUPC 2024 and CUET IUPC 2023.
- Finalist, CUET Tech Carnival 2020.

Selected Coursework

Algorithms and Data Structures; Database Systems; Computer Networks; Operating Systems; Software Engineering; Artificial Intelligence; Machine Learning; Information Security.